PostalMS

Team Meeting Minutes

CST2550 | The Leftovers | Middlesex University

Meeting 1

Date	Monday 3 March 2026
Time	2:00pm - 3:00pm
Location	Ritterman Building, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyur Bhai Patel, Forhad Hossain
Absent	None

Agenda

- Discuss the project requirements from the CST2550 brief
- Agree on roles and responsibilities for each team member
- Set up the GitHub repository structure
- Plan the first sprint tasks and deadlines

Discussion

The team met in the Ritterman Building to go through the CST2550 coursework brief in detail. The project was confirmed as a Postal Service Management System using C# WinForms and Microsoft SQL Server.

The team discussed the requirements including the need to implement custom data structures without using any standard template library collections. Three data structures were identified as most suitable: a Hash Table for fast parcel lookup using O(1) average search, a Binary Search Tree for sorted parcel display using O(log n) insertion, and a Queue for FIFO delivery processing using O(1) enqueue and dequeue.

Roles were agreed and assigned to each team member based on their skills and availability. The GitHub repository structure was planned with separate folders for design, docs, media, meetings and src.

Decisions Made

- Project name confirmed as PostalMS
- Application type confirmed as C# WinForms desktop application
- Database confirmed as Microsoft SQL Server 2022
- Three custom data structures selected: Hash Table, BST, Queue
- GitHub repository to be created and set up by the Team Leader
- Roles assigned: Durga (Team Leader and Developer), Meet (Secretary), Tapan (Developer), Forhad (Tester).

Action Points

- **Durga:** Create GitHub repository and push initial C# WinForms project by 6 March
- **Meet:** Set up meetings folder and begin documenting meeting minutes
- **Tapan:** Research BST recursive implementation approach
- **Forhad:** Research MS Test unit testing framework setup and Review the full coursework brief requirements carefully
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Next Meeting

Monday 10 March 2026 — Sheppard Library, Middlesex University

Meeting 2

Date	Monday 10 March 2026
Time	1:00pm - 2:30pm
Location	Sheppard Library, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyur Bhai Patel, Forhad Hossain.
Absent	None

Agenda:

- Review progress from Sprint 1
- Review the GitHub repository setup
- Discuss and agree on the database schema design
- Discuss the data structure pseudocode and time complexity

- Plan Sprint 2 tasks

Discussion:

The team met in the Sheppard Library to review Sprint 1 progress. Durga presented the initial GitHub repository with the folder structure created and the C# WinForms project set up.

The database schema was discussed in detail. The team agreed on three main tables: Users, Parcel and Delivery. The relationships between tables were reviewed and the ERD was sketched out. The stored procedures required for login, registration, parcel management and delivery management were identified.

Tapan presented his research on the BST implementation. The recursive approach for insert, search and delete was agreed upon. Durga presented the Hash Table design using chaining for collision resolution with a load factor of 0.75 triggering automatic resize.

The pseudocode approach was discussed. It was agreed that pseudocode would be written for all three data structures including Add, Search, Delete, Resize and Clear for the Hash Table, Insert, Search, InOrder, FindMin and Delete for the BST, and Enqueue, Dequeue, Peek, IsEmpty and ToList for the Queue.

Decisions Made:

- Database to have three tables: Users, Parcel and Delivery
- Hash Table to use chaining for collision resolution with load factor 0.75
- BST to use recursive insert, search and delete operations
- Queue to be implemented as a linked list with front and rear pointers
- SQL script to include stored procedures and sample data for testing
- Class diagram and ERD to be created in draw.io and saved to the design folder

Action Points:

- **Durga:** Implement CustomHashTable and CustomBST classes from scratch
- **Durga:** Write the complete SQL database script with all tables, views and stored procedures
- **Tapan:** Implement CustomQueue class from scratch
- **Meet:** Create class diagram and ERD in draw.io and push to design folder
- **Forhad:** Begin writing unit test cases for Hash Table operations and Begin writing unit test cases for Queue operations

Next Meeting

Monday 17 March 2026 — Ritterman Building, Middlesex University

Meeting 3

Date	Monday 17 March 2026
Time	2:00pm - 3:30pm
Location	Ritterman Building, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyurbhai Patel, Forhad Hossain.
Absent	None

Agenda

- **Review Sprint 2 progress and data structure implementations**
- **Discuss UI design and application navigation structure**
- **Review the database script and sample data**
- **Review initial unit test cases**
- **Plan Sprint 3 tasks**

Discussion

The team met in the Ritterman Building to review Sprint 2 progress. Durga demonstrated the working CustomHashTable and CustomBST implementations. Both data structures had been manually tested and were functioning correctly with chaining collision resolution and recursive tree operations.

Tapan demonstrated the CustomQueue implementation with a linked list using front and rear pointers. All three data structures were confirmed as working without using any standard template library collections.

The SQL database script was reviewed by the full team. All three tables had been created with the correct columns and relationships. Five sample parcels and deliveries had been added for the test user John Smith to allow full testing of the application.

The UI design was discussed at length. The team agreed on a red colour theme throughout the entire application. The navigation structure was confirmed: Home, Parcels, Deliveries, Help and Info. An offline AI assistant was proposed as a bonus feature for additional marks.

Forhad presented the initial unit test cases for the Hash Table covering add, search and delete. The MSTest project had been set up correctly referencing the main application project.

Decisions Made

- Red colour theme confirmed for the entire PostalMS application
- Navigation confirmed: Home, Parcels, Deliveries, Help and Info pages
- AI assistant to be implemented as an offline rule-based system requiring no internet
- Unit tests to cover all three data structures, pricing logic and tracking ID generation
- PDF report structure agreed based on coursework brief sections
- Pseudocode document to be written with full explanations for all three data structures

Action Points

- **Durga:** Implement all WinForms pages and connect all features to the SQL database
- **Durga:** Implement the offline AI assistant with parcel tracking and price estimation
- **Tapan:** Assist with WinForms form layouts and UI styling
- **Meet:** Write the complete pseudocode document for all three data structures
- **Forhad:** Complete all 59 unit test cases and run them to confirm all pass and Begin writing the PDF report introduction and design sections

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Next Meeting

Monday 24 March 2026 — Sheppard Library, Middlesex University

PostalMS — Meeting Minutes

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Meeting 4

Date	Monday 24 March 2026
Time	1:00pm - 3:00pm
Location	Sheppard Library, Middlesex University
Attendees	Durga Prasad Avala, Meet Sanjay Mande, Tapan Keyurbhai Patel, Forhad Hossain.

Absent	None

Agenda

- Final review of all completed features and source code
- Review unit test results and confirm all 59 tests pass
- Review report and pseudocode document progress
- Plan the video demonstration
- Confirm final submission checklist

Discussion

The team held a final review meeting in the Sheppard Library. Durga demonstrated the complete PostalMS application running with all features working including the login and registration system, parcel sending with mail and package options, domestic and international shipping, real-time delivery tracking, refund requests, user profile management with photo upload, and the offline AI assistant.

Forhad presented the completed unit test suite. All 59 tests were run live during the meeting and all passed successfully in 66 ms with 0 failures and 0 skipped. The tests covered the Hash Table (12 tests), BST (13 tests), Queue (10 tests), price calculation (9 tests), tracking ID validation (6 tests) and integration between all three data structures (5 tests).

Meet presented the completed pseudocode document covering all three data structures with full explanations, step by step annotations and time complexity analysis tables. The document was confirmed as ready to upload to the design folder.

The team discussed the video demonstration plan. It was agreed the video would show the full application running, explain how the data structures work within the system, and demonstrate the unit tests passing. The video must be under 6 minutes as required by the coursework brief.

The final submission checklist was reviewed. All outstanding items were identified and deadlines assigned to each team member.

Decisions Made

- All source code to be pushed to GitHub by 7 April 2026
- PDF report to be completed and uploaded by 8 April 2026
- Video demonstration to be recorded and uploaded by 9 April 2026
- Final submission .txt file to be submitted before 10 April 2026 at 5:30pm
- No changes to be made to the repository after the submission deadline
- All 4 markers to be invited as collaborators to the GitHub repository

Action Points

- **Durga:** Final code review and push all source files and test project to GitHub
- **Durga:** Invite all 4 markers to the GitHub repository as collaborators

- **Meet:** Complete and upload the final PDF report to the docs folder
- **Tapan:** Record the MP4 video demonstration and upload to the media folder
- **Forhad:** Upload unit testing report to the docs folder and Final check of all repo folders to confirm everything is in place

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Next Meeting

No further meetings scheduled. Deadline: Friday 10 April 2026 at 5:30pm.